



Introduction to Computer Vision

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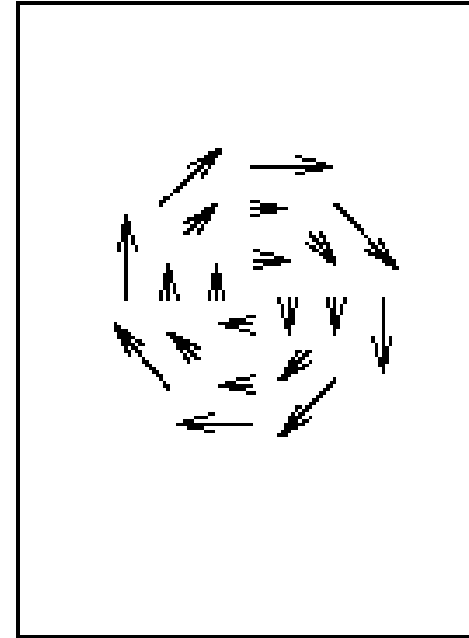
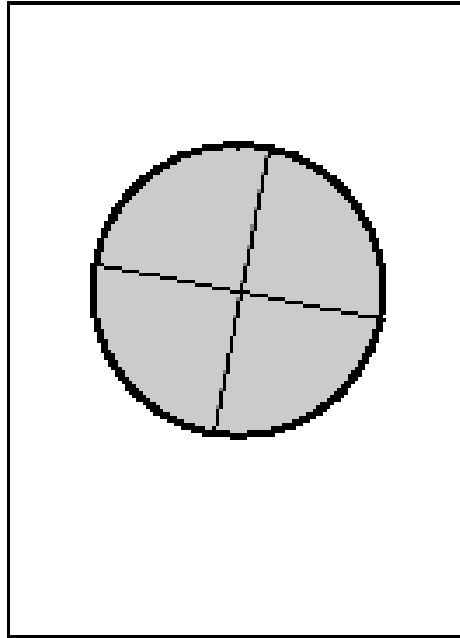
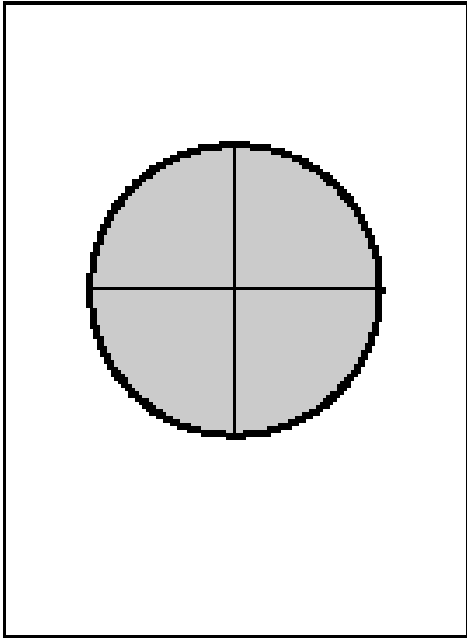
Lecture 8

Learning Outcomes

- Optical Flow



Optical Flow



Fundamental Equations of Computer Vision

1. Image Filtering

$$h[m,n] = \sum_{k,l} f[k,l] I[m+k, n+l]$$

2. Optical Flow

$$I(x, y, t) = I(x + \Delta x, y + \Delta y, t + \Delta t)$$

3. Camera Geometry

$$\mathbf{x} = \mathbf{K}[\mathbf{R} \ \mathbf{t}] \mathbf{X} \qquad \mathbf{x}^T \mathbf{F} \mathbf{x}' = 0$$

4. Machine Learning

$$\arg \min_{\mathbf{S}} \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2. \qquad y = \varphi\left(\sum_{i=1}^n w_i x_i + b\right) = \varphi(\mathbf{w}^T \mathbf{x} + b)$$

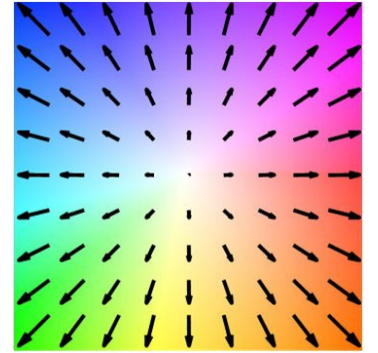
Optical Flow: Dense Correspondence Over Time



Input [Liu *et al.* CVPR'08]



Optical flow (2D motion vector)



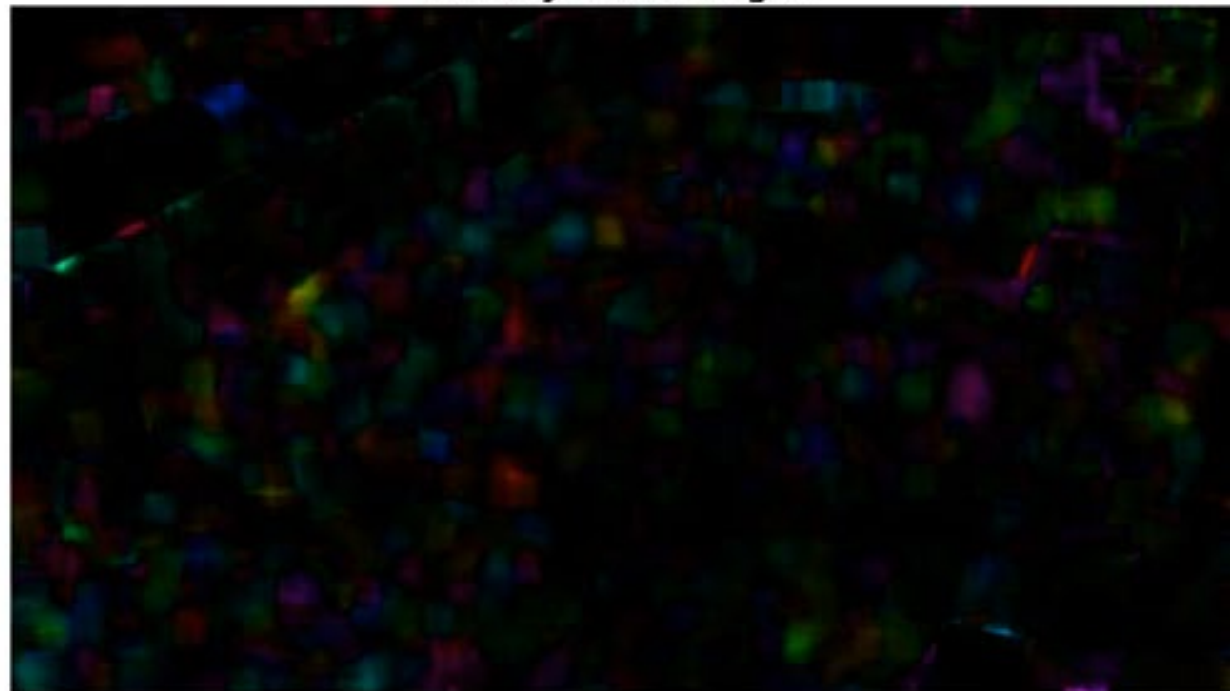
Color key
[Baker *et al.* IJCV'11]

Optical Flow to find the Walk Direction of people in a Video with OpenCV-python

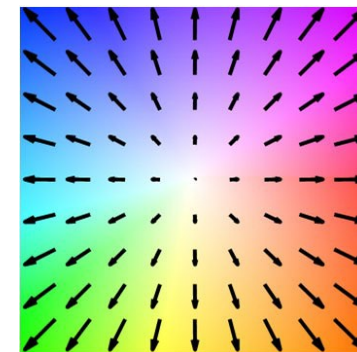
input



Velocity Vector Angles



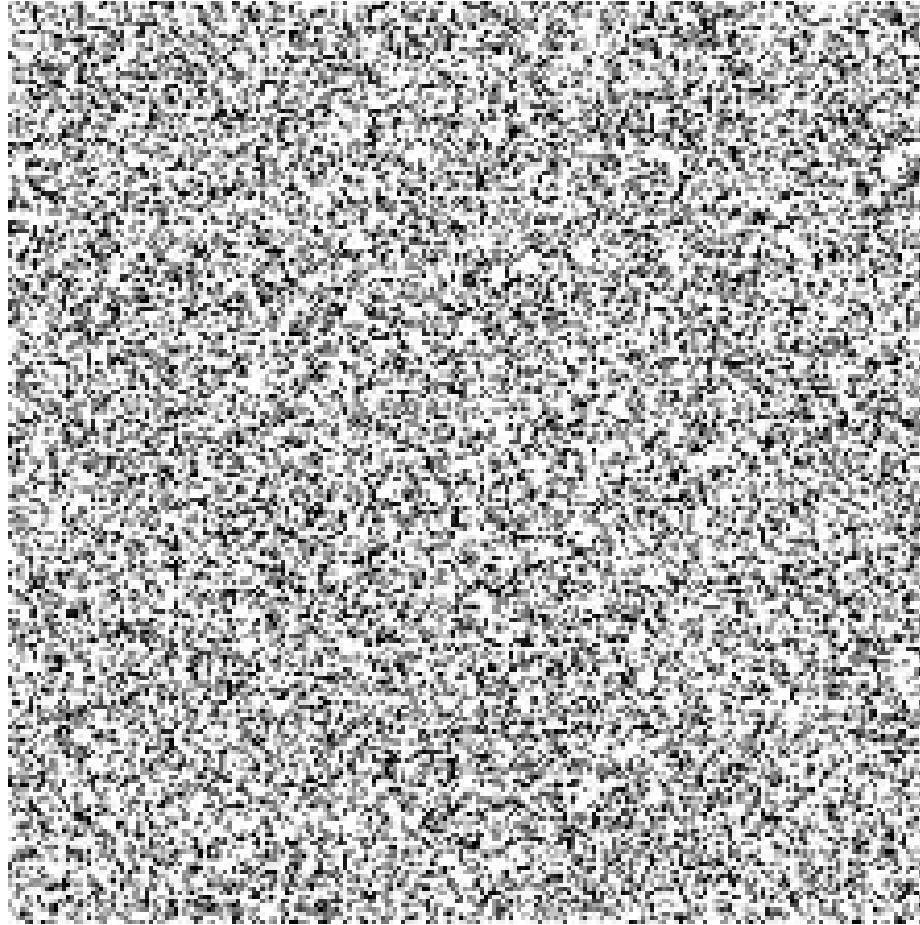
https://www.youtube.com/watch?v=e_TeY6QRp4c



Color key

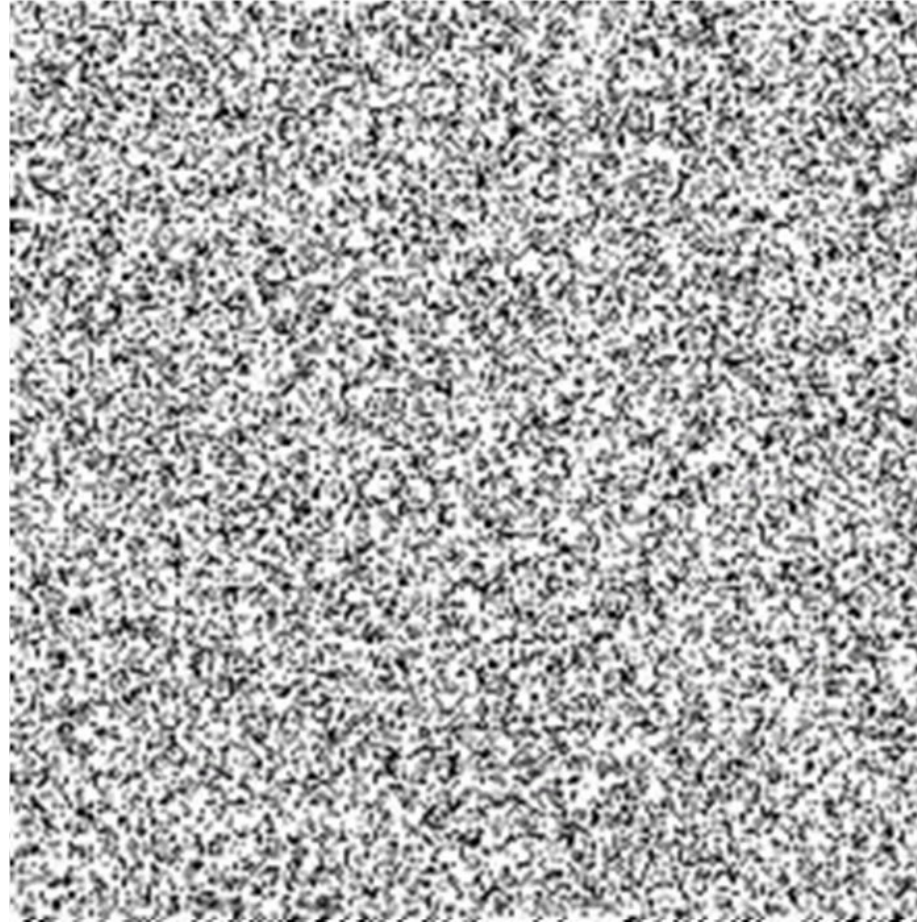
Correspondence and perceptual organization

Sometimes correspondence/motion is the only cue



Correspondence and perceptual organization

Sometimes correspondence/motion is the only cue





<https://www.youtube.com/watch?v=mNrqvcS0oI0>

Applications of optical flow



Video Superresolution
[Liu & Sun CVPR 2011,
TPAMI 2014]

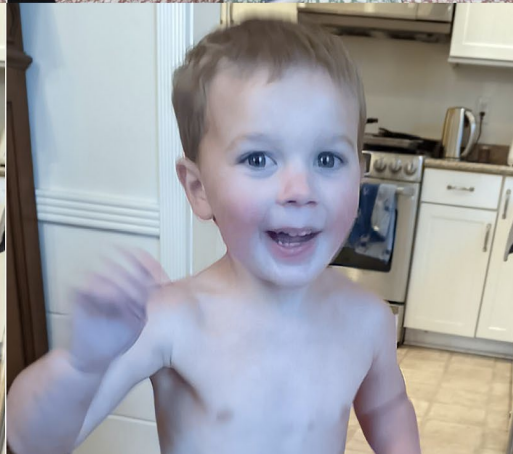
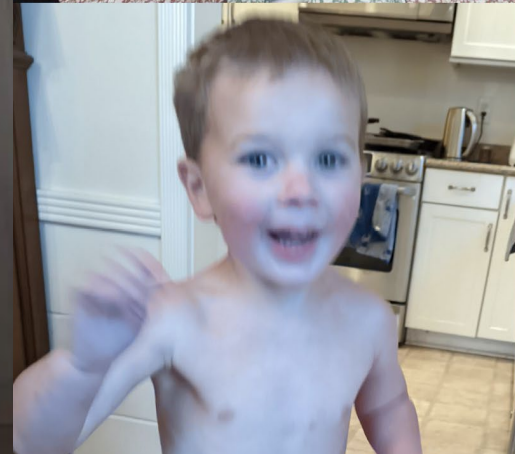
Applications of optical flow



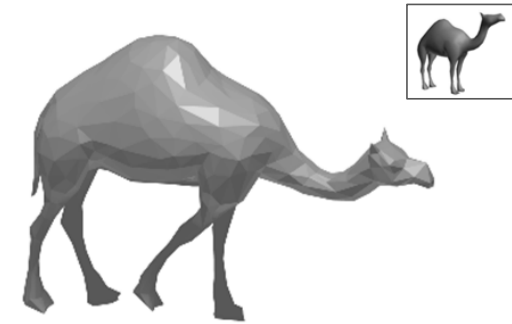
<https://youtu.be/MjViy6kyiqs?si=zXenEX606VJkEaQF>

Super SloMo [Jiang, Sun, *et al.* CVPR 2018 Spotlight] Incorporated into **NVIDIA NGX** SDK for the Turing GPU.

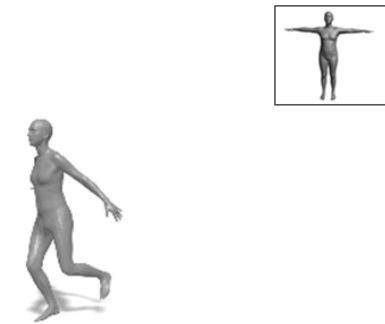
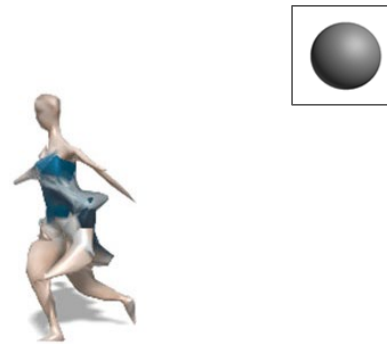
Face Unblur for Pixel 6



Articulated shape reconstruction from a monocular video



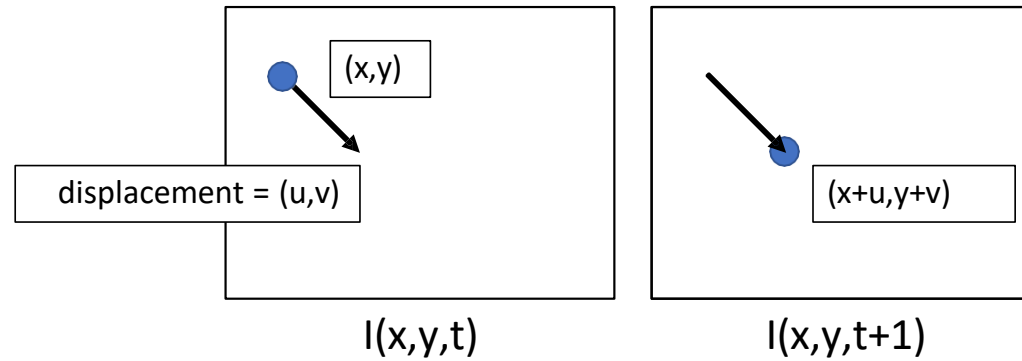
A-CSM: Kulkarni et al. CVPR 2020



VIBE: Kocabas et al. CVPR 2020

Challenge: Solving non-rigid 3D shape from 2D measurements without template or category prior is highly *under-constrained*

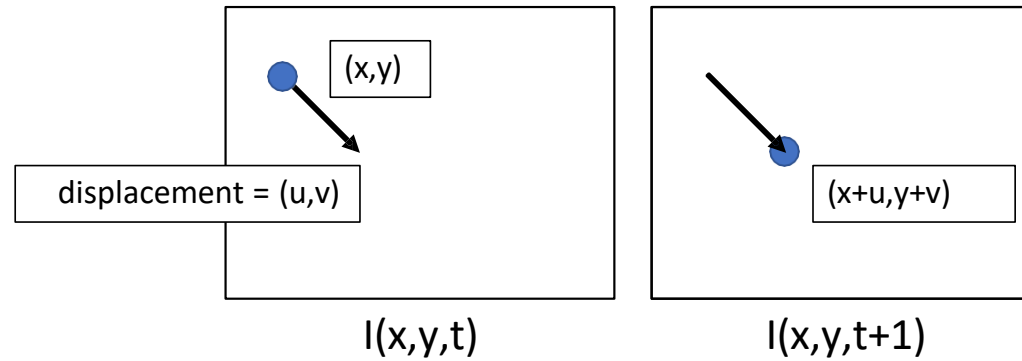
Optical Flow



Brightness
constancy:

$$I(x, y, t) = I(x + u, y + v, t + 1)$$

Optical Flow



Brightness
constancy:

$$I(x, y, t) = I(x + u, y + v, t + 1)$$

Recall Taylor
Expansion:

$$I(x + u, y + v, t + 1) = I(x, y, t) + I_x u + I_y v + I_t \dots$$

Optical Flow Equation

$$\begin{aligned} I(x + u, y + v, t + 1) &= I(x, y, t) \\ 0 &= I(x + u, y + v, t + 1) - I(x, y, t) && \text{Taylor} \\ &\approx I(x, y, t) + I_t + I_x u + I_y v - I(x, y, t) && \text{Expansion} \\ &= I_t + I_x u + I_y v \\ &= I_t + \nabla I \cdot [u, v] \end{aligned}$$

When is this approximation **bad**?
u or v big

Optical Flow Equation

Brightness constancy equation

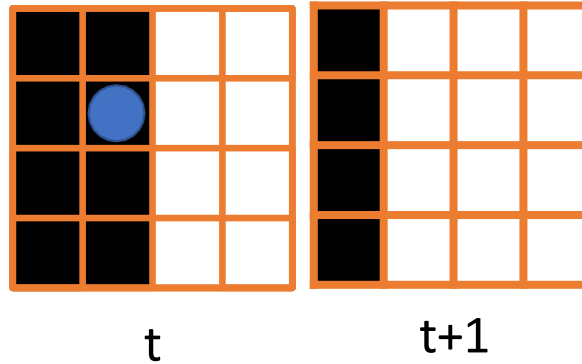
$$I_x u + I_y v + I_t = 0$$

What do static image gradients have to do with motion estimation?



Brightness Constancy Example

$$I_x u + I_y v + I_t = 0$$



$$I_t = 1 - 0 = 1$$

@  $I_y = 0$

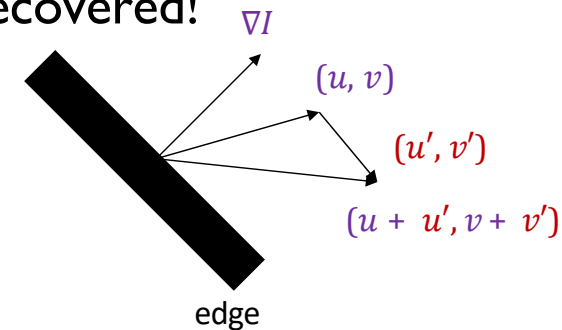
$$I_x = 1 - 0 = 1 \quad \text{--> } u + 1 = 0 \text{ --> } u = -1$$

What's u? What's v?

The Brightness Constancy Constraint

$$I_x u + I_y v + I_t = 0$$

- Given the gradients I_x , I_y and I_t , can we uniquely recover the motion (u, v) ?
 - One equation, two unknowns
 - Suppose (u, v) satisfies the constraint: $\nabla I \cdot (u, v) + I_t = 0$
 - Then $\nabla I \cdot (u + u', v + v') + I_t = 0$ for any (u', v') s. t. $\nabla I \cdot (u', v') = 0$
 - Interpretation: the component of the flow perpendicular to the gradient (i.e., parallel to the edge) cannot be recovered!



Solving Ambiguity – Lucas Kanade

- 2 unknowns [u,v], 1 eqn per pixel How do we get more equations?
- Assume **spatial coherence**: pixel's neighbors have move together / have same [u,v]
- 5x5 window gives 25 new equations

$$I_t + I_x u + I_y v = 0$$
$$\begin{bmatrix} I_x(p_1) & I_y(p_1) \\ \vdots & \vdots \\ I_x(p_{25}) & I_y(p_{25}) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} I_t(p_1) \\ \vdots \\ I_t(p_{25}) \end{bmatrix}$$

B. Lucas and T. Kanade. An iterative image registration technique with an application to stereo vision. In *Proceedings of the International Joint Conference on Artificial Intelligence*, pp. 674–679, 1981.

Solving for u,v

$$\begin{bmatrix} I_x(p_1) & I_y(p_1) \\ \vdots & \vdots \\ I_x(p_{25}) & I_y(p_{25}) \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} I_t(p_1) \\ \vdots \\ I_t(p_{25}) \end{bmatrix} \quad \begin{matrix} \mathbf{A} & \mathbf{d} = \mathbf{b} \\ 25 \times 2 & 2 \times 1 & 25 \times 1 \end{matrix}$$

What's the solution?

$$(\mathbf{A}^T \mathbf{A}) \mathbf{d} = \mathbf{A}^T \mathbf{b} \rightarrow \mathbf{d} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

Intuitively, need to solve (sum over pixels in window)

$$\underbrace{\begin{bmatrix} \sum I_x I_x & \sum I_x I_y \\ \sum I_x I_y & \sum I_y I_y \end{bmatrix}}_{\mathbf{A}^T \mathbf{A}} \begin{bmatrix} u \\ v \end{bmatrix} = - \underbrace{\begin{bmatrix} \sum I_x I_t \\ \sum I_y I_t \end{bmatrix}}_{\mathbf{A}^T \mathbf{b}}$$

Challenges for traditional methods



Large motion, motion blur



Textureless regions

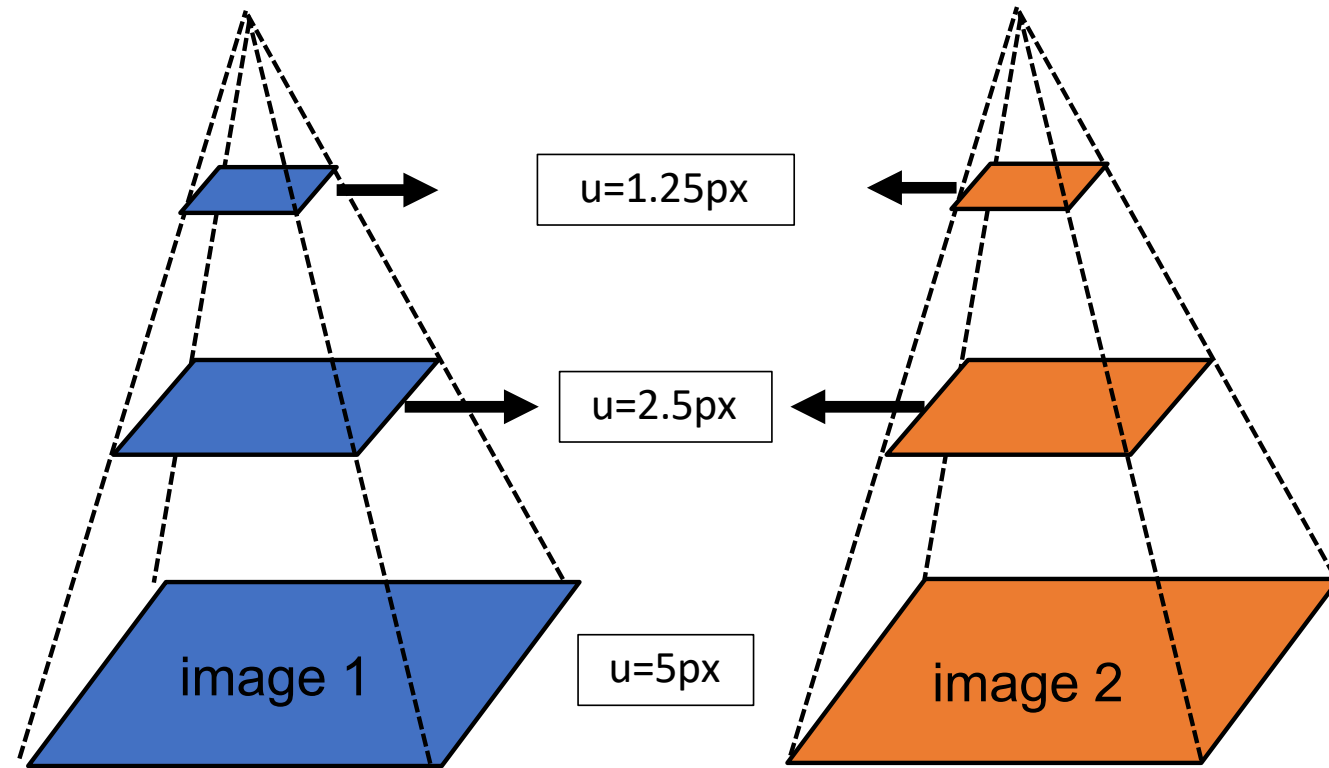


Occlusions



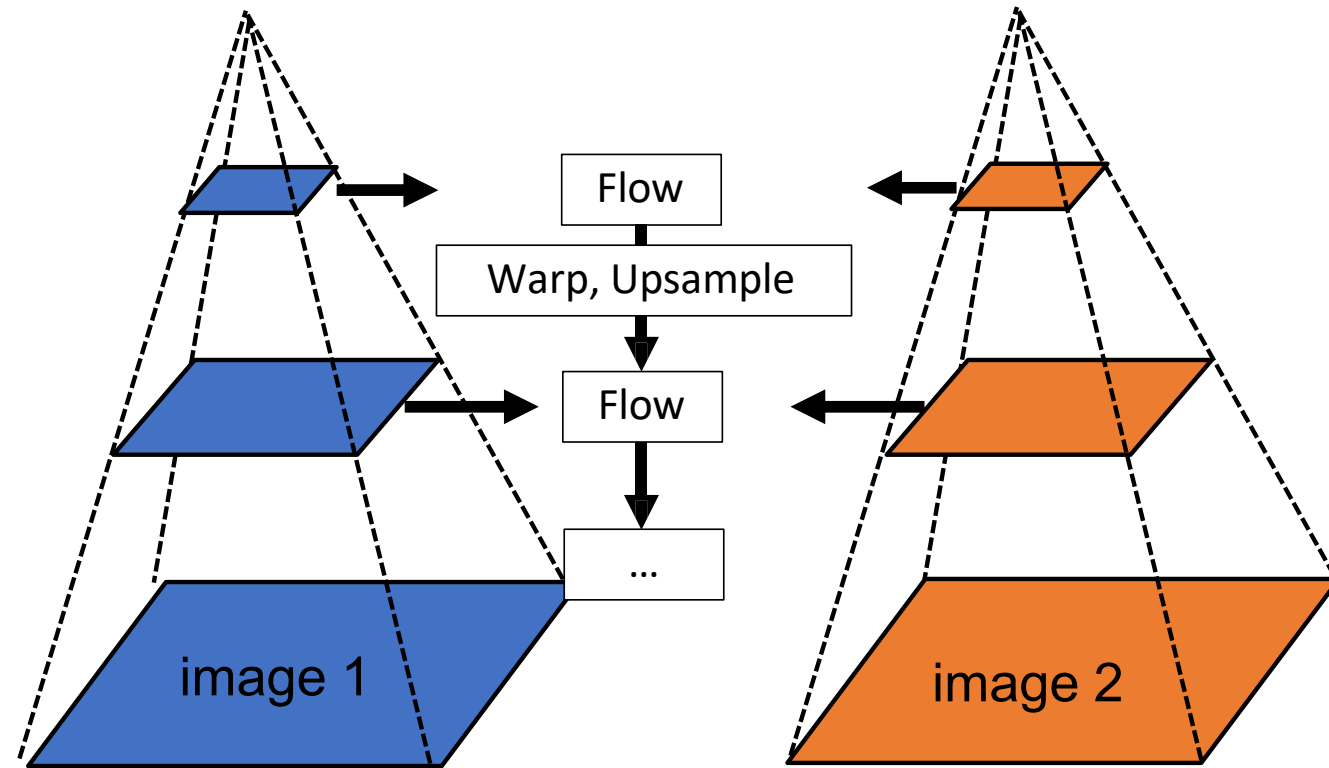
Lighting changes, noise ...

Coarse-to-fine optical flow estimation

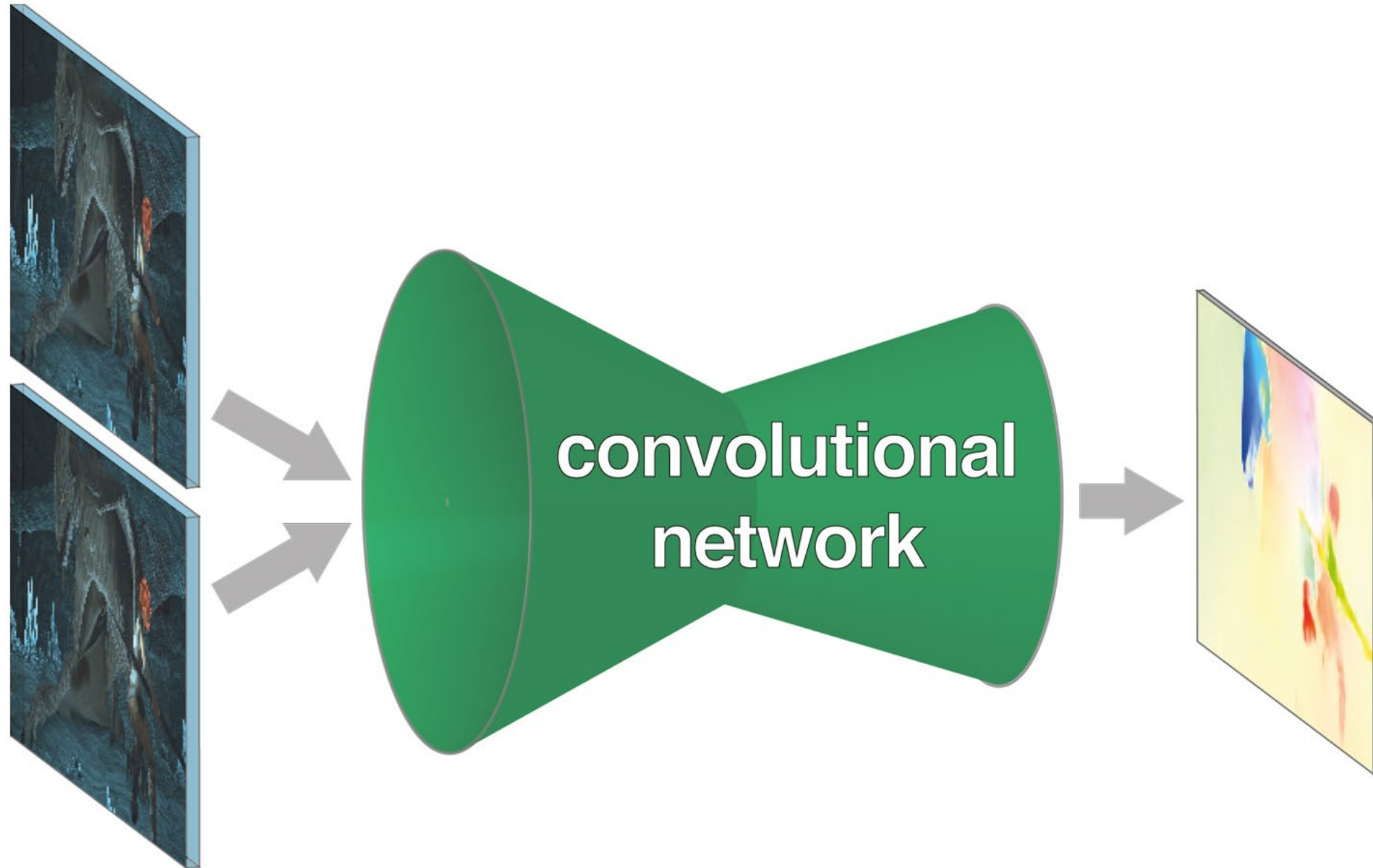


Typically called Gaussian Pyramid

Coarse-to-fine optical flow estimation



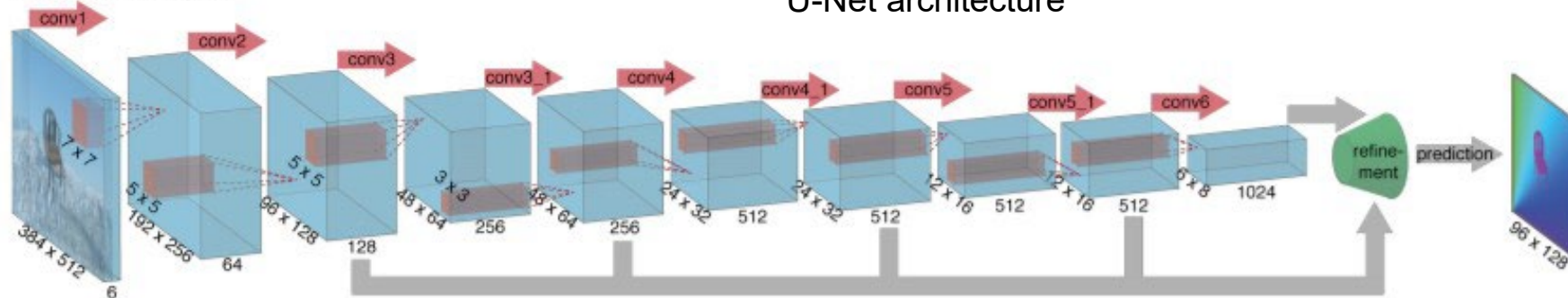
Typical DL Pipeline for Optical Flow



FlowNetS: mapping from images to flow

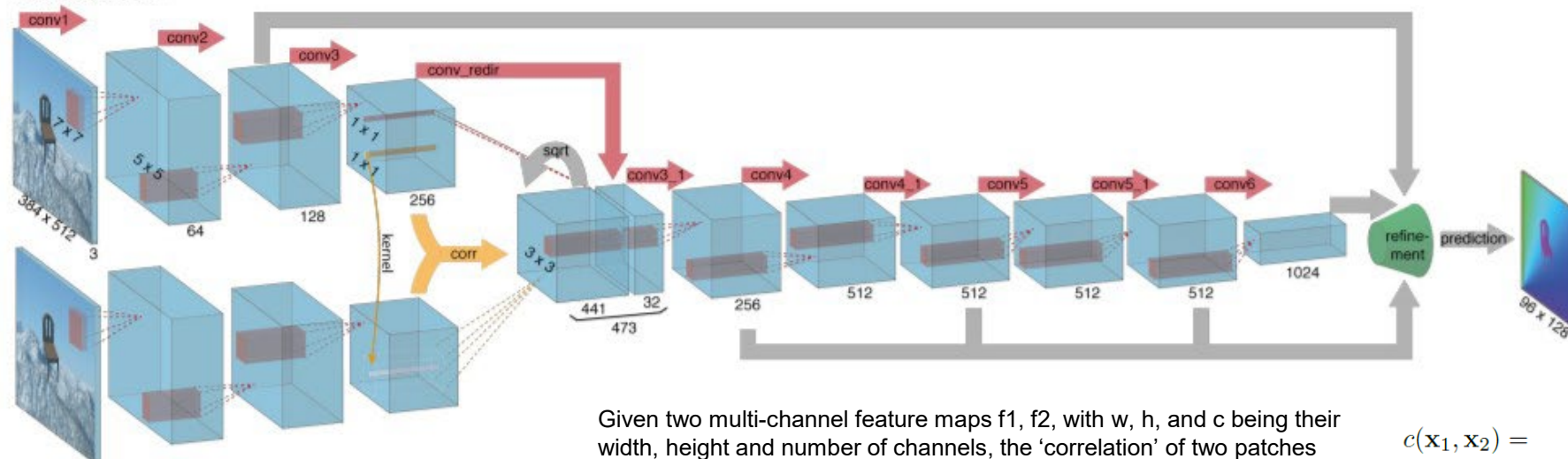
[Dosovitskiy *et al.* ICCV'15]

FlowNetSimple



U-Net architecture

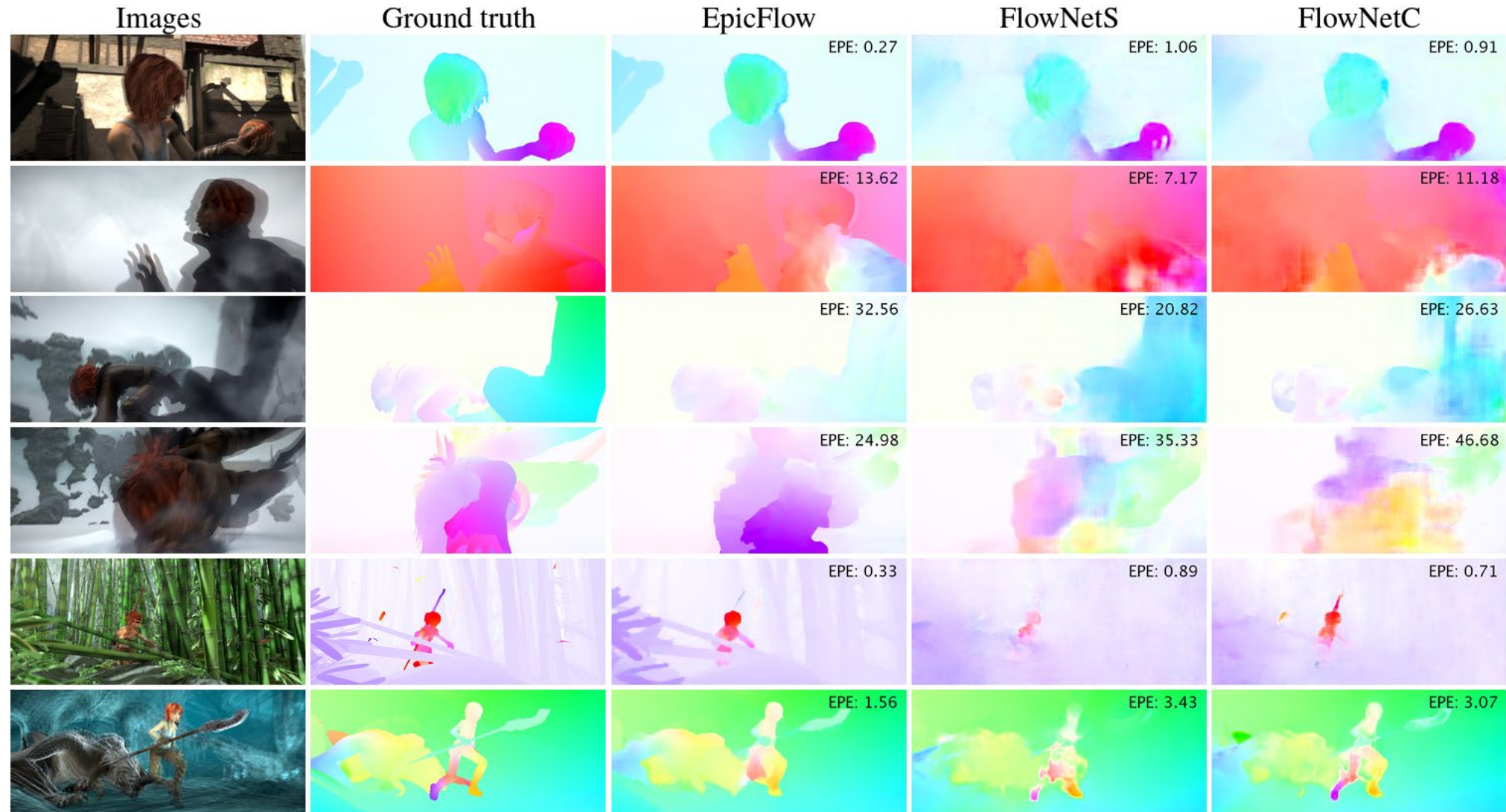
FlowNetCorr



Given two multi-channel feature maps f_1 , f_2 , with w , h , and c being their width, height and number of channels, the 'correlation' of two patches centered at x_1 in the first map and x_2 in the second map is then defined as:

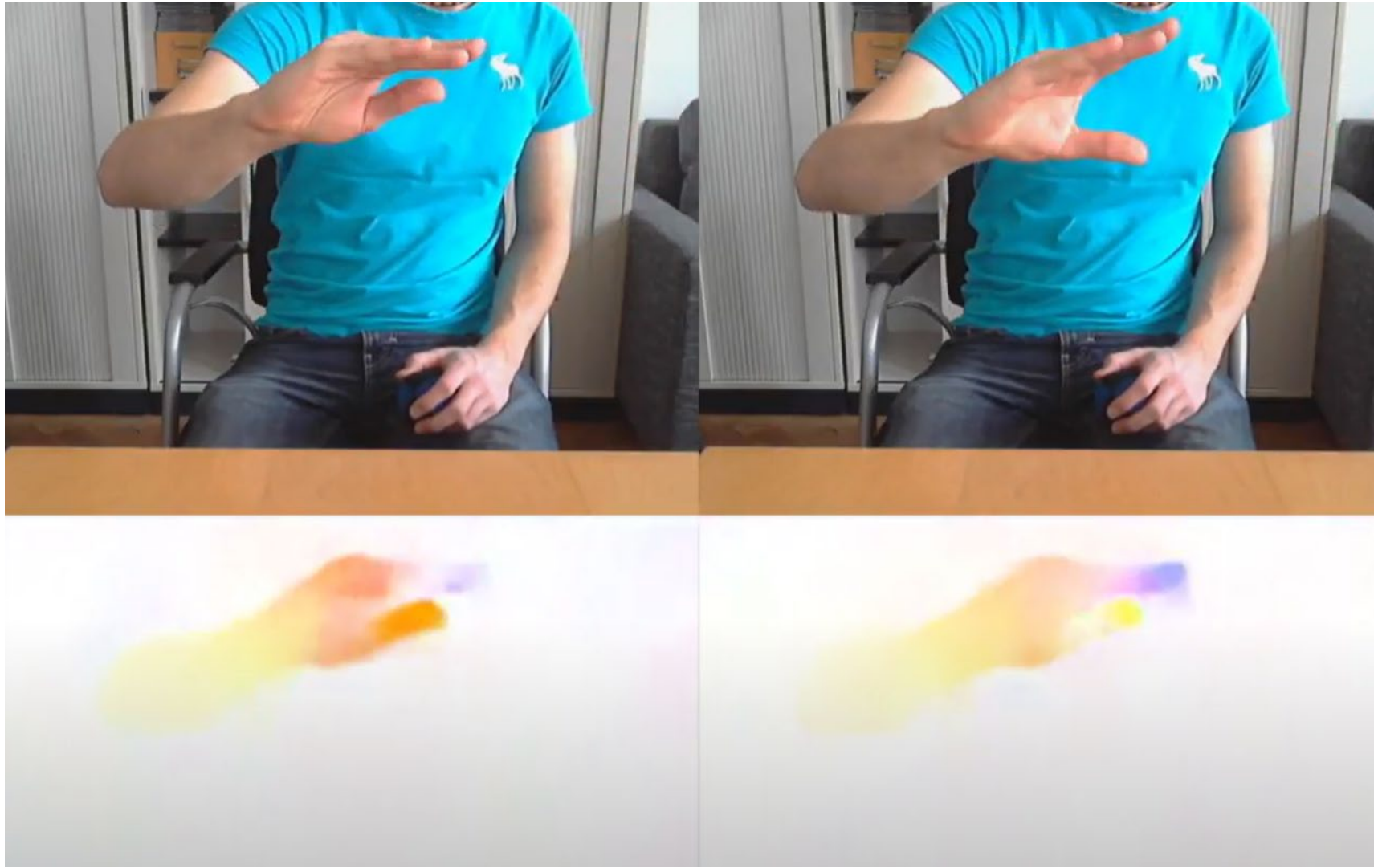
$$c(x_1, x_2) = \sum_{o \in [-k, k] \times [-k, k]} \langle f_1(x_1 + o), f_2(x_2 + o) \rangle$$

FlowNetS: mapping from images to flow



EPE: endpoint error

FlowNetS: mapping from images to flow



https://youtu.be/k_wkDLJ8lJE?si=wJj1RePbw-NA75lU